

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T44 · Circles, Arcs and Sectors

Questions

Student book · questions with writing space

■ 25 pages

FAMILY MAP

Circles Arcs Sectors

F01	Use arc and sector measures	5 Q 14 marks
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F02	Resolve compound and shaded circle regions	14 Q 66 marks
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F03	Recognise parts of a circle	1 Q 4 marks
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F04	Use full-circle and semicircle measures	1 Q 5 marks
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Working Unit 08

4MA1-1H Q11

UNIT 1
2 marks

- 11 $OABC$ is a sector of a circle, centre O

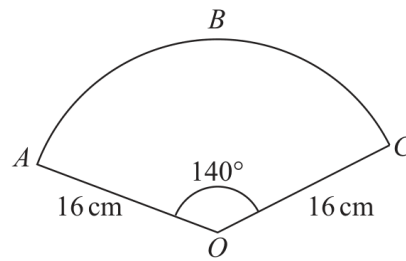


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16 \text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

Question 14

4MA1-1H Q20

MODULAR UNIT 1

4 marks

20 The diagram shows a sector $OABC$ of a circle centre O

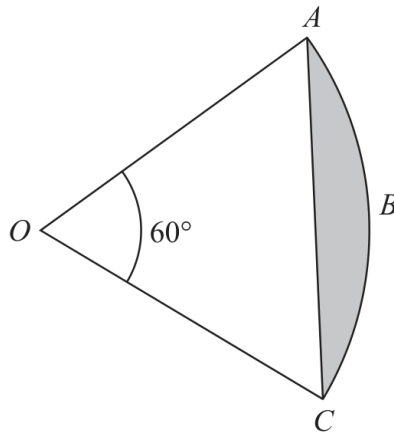


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

..... cm

Question 08

4MA1-1H Q5 demand

MODULAR UNIT 1

3 marks

- 5 The diagram shows a shaded shape made from three identical semicircles, *X*, *Y* and *Z*

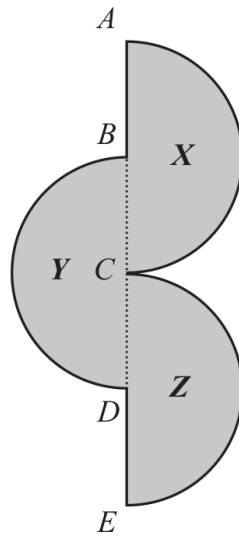


Diagram **NOT**
accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle *X* and *B* is the centre of semicircle *X*

BD is the diameter of semicircle *Y* and *C* is the centre of semicircle *Y*

CE is the diameter of semicircle *Z* and *D* is the centre of semicircle *Z*

$$AC = BD = CE = 20 \text{ cm}$$

Work out the perimeter of the shaded shape.

Give your answer correct to the nearest whole number.

..... cm

Question 30

4MA1-2H Q20 whole

MODULAR UNIT 1

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm. A circle is drawn inside the triangle.

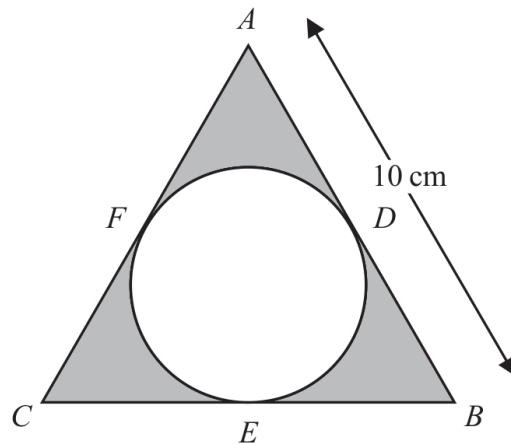


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

..... cm^2

Question 11

4MA1-1H Q22

MODULAR UNIT 1

6 marks

- 22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

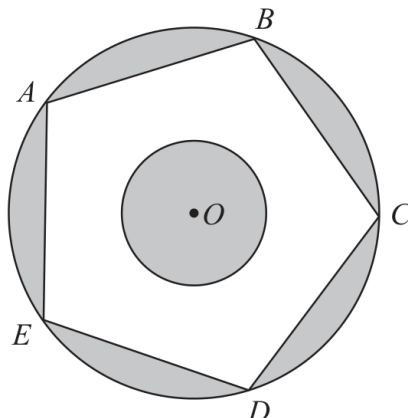


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Question 12

4MA1-1H Q16 M01

MODULAR UNIT 1

4 marks

16 $OAPB$ is a sector of a circle, centre O

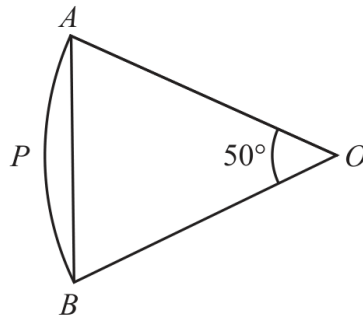


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

Question 28

4MA1-2H Q25 M01

MODULAR UNIT 1

5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

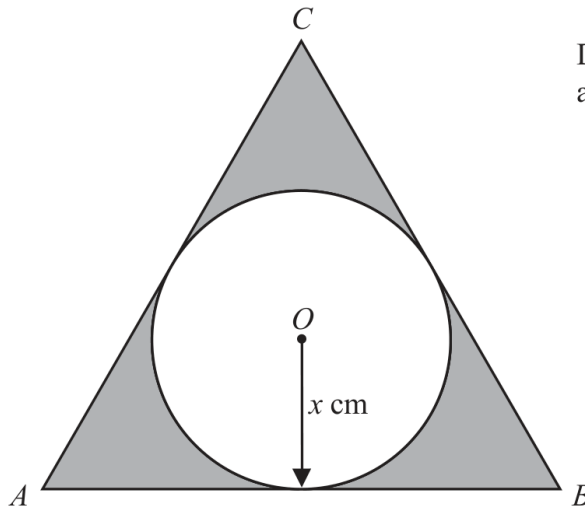


Diagram **NOT**
accurately drawn

AB, BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

4MA1-2H Q25 M01

5 marks

$n =$

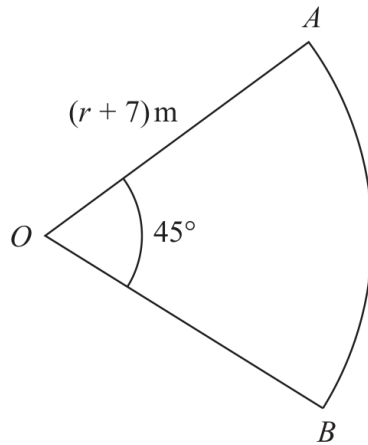
Question 13

4MA1-1H Q19 Q19

MODULAR UNIT 1

5 marks

19

Diagram **NOT**
accurately drawn

OAB is a sector **S** of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45^\circ$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r

Show your working clearly.

Question 19 continued.

Question 27

4MA1-2H Q20 Q20

MODULAR UNIT 1

4 marks

- 20 The diagram shows four identical circles drawn inside a square.

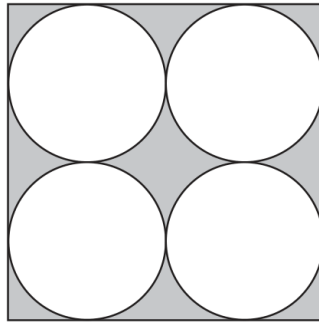


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

Question 14

4MA1-1H Q16

MODULAR UNIT 1

4 marks

16 The diagram shows a circle with centre O

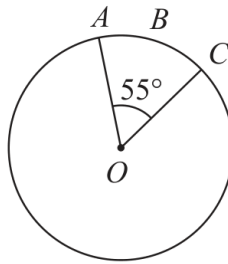


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

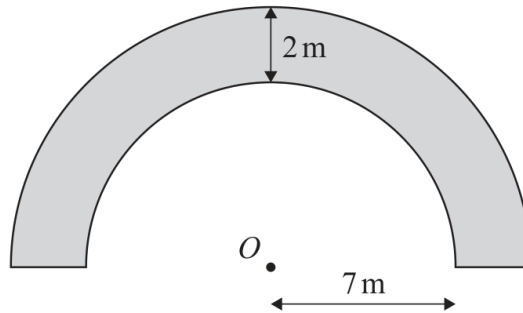
Give your answer correct to one decimal place.

..... cm^2

-

.....cm

- Diagram **NOT**
accurately drawn



Give your answer correct to one decimal place.

..... m

A circle with center O is shown. A triangle ABC is inscribed in the circle, with the right angle at vertex B . The side AB is labeled 8 cm and the side BC is labeled 15 cm . The region of the circle outside the triangle is shaded gray.

.....cm²

The diagram shows a sector OAC with center O and angle 50° . Point D is on the radius OC such that $OD = 6$ cm and $DC = x$ cm. A line segment AD is drawn. A line segment AB is drawn parallel to AD , where B is a point on the arc AC . The region $OADB$ is shaded.

Diagram **NOT**
accurately drawn

Give your answer correct to 3 significant figures.

A diagram of a circular sector. The two straight sides (radii) are labeled 7 cm . The central angle is labeled 50° .

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

- 25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

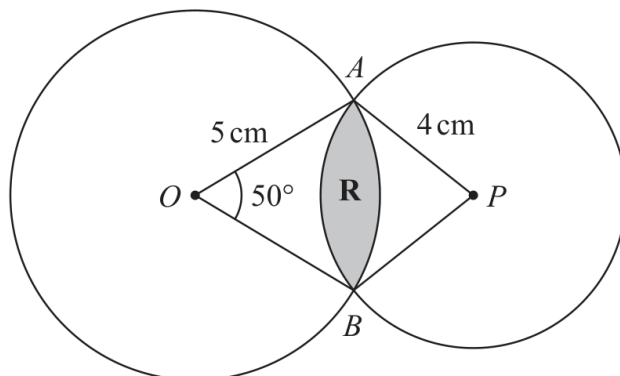


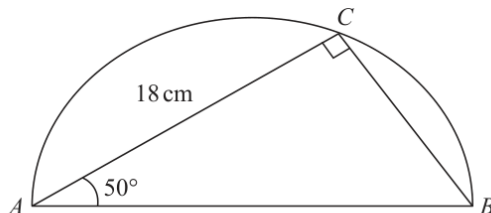
Diagram **NOT**
accurately drawn

One of the circles has centre O and radius 5 cm .
The other circle has centre P and radius 4 cm .
Angle $AOB = 50^\circ$

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.

..... cm^2

Diagram **NOT**
accurately drawn



Work out the perimeter of the semicircle.
Give your answer correct to 2 significant figures.

..... cm

(Total for Question 8 is 5 marks)

A circle with center O . A sector OAC is shown with radius $OA = 4.8$ cm and central angle $\angle AOC = 72^\circ$. The region between the arc AC and the chord AC is shaded gray and labeled B .

$OABC$ is a sector of the circle, centre O

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes. Give your answer in cm^3 correct to 3 significant figures.

..... cm^3

(Total for Question 21 is 5 marks)

A diagram of a circular sector AOB. The center is labeled O. The radius OB is labeled 12 cm. The angle AOB is labeled 75°. The arc AB is shown as a curved line connecting points A and B.

Work out the area of the sector.
Give your answer correct to 3 significant figures.

.....cm²

(Total for Question 15 is 2 marks)

- 6 A large circle with centre O contains 3 identical small circles. Each circle touches two other circles.

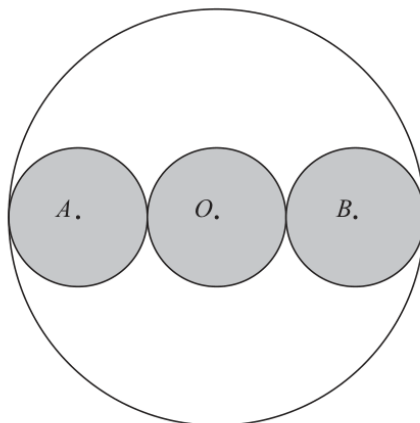


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... cm^2

Question 11

4MA1-1H Q10 Q10(a), Q10(b)

MODULAR UNIT 2

4 marks

- 10 Two circles, C_1 and C_2 , are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates $(-1, 3)$ and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates $(7, 18)$ and the radius of C_2 is 6 cm.

- (a) Work out the distance between the centre of C_1 and the centre of C_2

..... cm

(3)

- (b) Explain why circle C_1 intersects circle C_2